

Microsoft ADC Cybersecurity Skilling Program

Week 8 Assignment 14

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SECURE COMPUTE

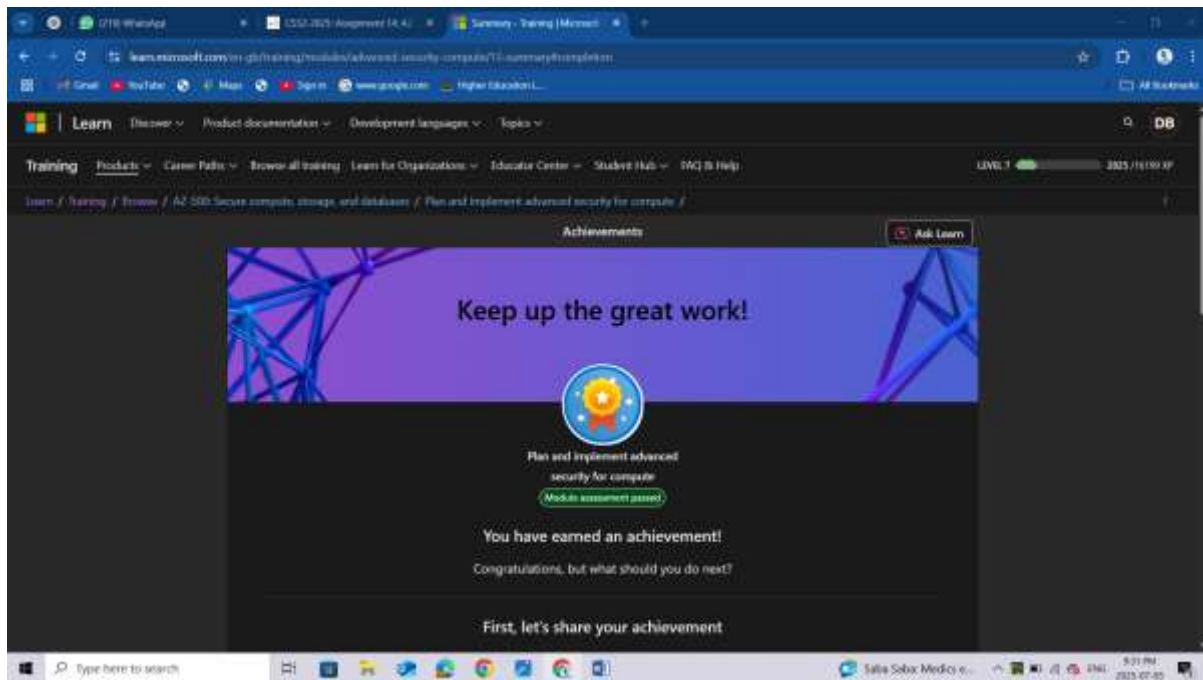
Introduction

In this module, I will **focus on planning and implementing advanced security for compute resources** in Microsoft Azure. Compute resources—such as virtual machines (VMs), containers, and app services—**play a central role** in cloud workloads, and securing them **remains critical** to maintaining a strong and resilient security posture. Throughout the module, I will **explore** key strategies and tools that **help harden** Azure-based compute environments. I will **utilize** Microsoft Defender for Cloud to assess, monitor, and **respond to threats** targeting compute workloads. I will also **apply** just-in-time (JIT) VM access, **configure** endpoint protection, and **enable** adaptive application controls to help **reduce the attack surface** and prevent unauthorized access. Additionally, I will **work with** Azure Policy to enforce consistent security configurations, **implement** disk encryption using Azure-managed keys, and **leverage** secure boot and virtual Trusted Platform Module (vTPM) to provide hardware-level protection. These practices **lay the foundation** for implementing a defense-in-depth strategy that will help me secure Azure compute workloads both now and in the future.

Tasks Completed

Plan and implement advanced security for compute

<https://learn.microsoft.com/api/achievements/share/en-us/DeborahBinyanya-1253/H7JZZZ78?sharingId=BBBCB3F098199A41>



Conclusion

After completing this module, I have developed a comprehensive understanding of how to secure Azure compute resources using advanced, layered security techniques. I learnt how to monitor the security state of VMs and other compute workloads using **Microsoft Defender for Cloud**, and how to proactively reduce exposure through **just-in-time access**, **network hardening**, and **application control policies**. Additionally, I now understand how to implement **disk encryption**, configure **antimalware solutions**, and enforce security baselines using **Azure Policy** to ensure all compute resources comply with organizational standards. Features such as **secure boot** and **vTPM** offer further assurance by validating the integrity of the VM operating system at startup. This module has equipped me with the skills needed to design and maintain a secure compute environment in Azure—one that is resilient to modern threats, aligned with best practices, and tailored to meet enterprise security and compliance requirements.